

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|---------------------------------|--|
| 1. Function _____ | (A) The set of every output the function actually produces. |
| 2. Domain _____ | (B) The simplest form of a family, before any shift or stretch. |
| 3. Range _____ | (C) A rule that sends each input to <i>exactly one</i> output. |
| 4. Average rate of change _____ | (D) The function that reverses the input-output pairs of f . |
| 5. Concave up _____ | (E) Feeding the output of one function into another: $f(g(x))$. |
| 6. Parent function _____ | (F) The slope of the secant line, $\frac{f(b) - f(a)}{b - a}$. |
| 7. Composition _____ | (G) The set of every input the function is allowed to take. |
| 8. Inverse function _____ | (H) Bending upward — the rate of change is increasing. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

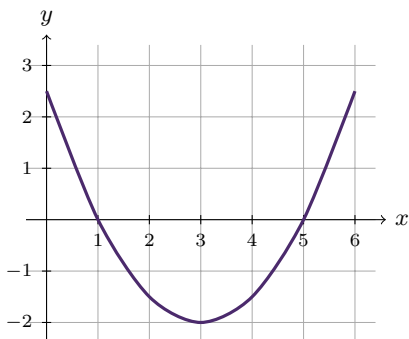
- A relation fails to be a function exactly when (A) two inputs share an output (B) one input has two outputs
(C) the graph crosses the x -axis (D) the domain is restricted
- If a function is decreasing and concave up on an interval, its output is (A) falling faster and faster (B) falling, but more and more slowly
(C) rising more and more slowly (D) changing at a constant rate
- A table of f has equally spaced inputs and **constant second** differences. The best model is (A) linear (B) quadratic (C) exponential (D) none of these
- The graph of $g(x) = f(x + 3)$ is the graph of f shifted (A) 3 right (B) 3 left (C) 3 up (D) 3 down
- If $f(x) = x^2$ and $g(x) = x - 1$, then $(f \circ g)(4)$ equals (A) 9 (B) 15 (C) 16 (D) 3
- A function has an inverse that is also a function exactly when the function is (A) increasing somewhere (B) one-to-one (C) concave up (D) defined at 0

Part C — Short Answer & Computation (30 pts)

Show your work.

1. Let $f(x) = 2x^2 - 5$. Compute **(a)** $f(3)$, **(b)** $f(-2)$, **(c)** the value of x (there are two) for which $f(x) = 3$.

2. The graph of $y = f(x)$ is shown below. **(a)** On what interval is f decreasing? **(b)** Is f concave up or concave down on the whole picture? **(c)** List the zeros of f . **(d)** What is $f(3)$?



3. A tank is draining. The volume V (gallons) after t minutes is given below.

t (min)	0	4	8	12
V (gal)	180	148	124	108

(a) Find the average rate of change of V from $t = 0$ to $t = 8$. **(b)** Give its units and say in one sentence what it means for the tank. **(c)** Is the tank draining faster or slower as time goes on? Justify with the numbers.

4. The table below has equally spaced inputs. Compute the first differences and the second differences, then name the function type (linear, quadratic, or neither).

x	0	1	2	3	4
$g(x)$	2	5	12	23	38

5. The parent function is $f(x) = \sqrt{x}$, with domain $[0, \infty)$ and range $[0, \infty)$. Let $g(x) = -2f(x - 1) + 4$. **(a)** Describe every transformation, in order. **(b)** State the domain and range of g .

6. Let $f(x) = 3x + 1$ and $g(x) = x^2 - 2$. **(a)** Compute $f(g(2))$. **(b)** Compute $g(f(2))$. **(c)** Write a formula for $(f \circ g)(x)$. **(d)** What do (a) and (b) show about the order of composition?
7. Let $f(x) = \frac{x-5}{2}$. **(a)** Find $f^{-1}(x)$ using the swap-and-solve method. **(b)** Verify your answer by computing $f(f^{-1}(x))$.

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. A rectangular dog run is built against a barn wall, so only **three** sides need fencing. The builder has 60 feet of fence. Let x be the length of each of the two sides perpendicular to the barn. **(a)** Write the area A as a function of x . **(b)** What type of function is A , and how do you know? **(c)** State the domain restriction that the context forces, and explain why. **(d)** State one assumption this model rests on.
2. Let $f(x) = x^2$ with domain all real numbers. **(a)** Explain why f has no inverse function on that domain. **(b)** Give a domain restriction that fixes the problem, and write the resulting inverse. **(c)** Explain how the graph of a function and the graph of its inverse are related.